Shallow crustal shear velocity of ∼40 Ma Pacific lithosphere from seafloor compliance at the Young ORCA array

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SUMMARY

We apply the compliance method of Aminian et al. (2025), now implemented end to end in ComPy 2.4.0, to a second broad-band ocean-bottom network: the 30-station Young ORCA array (FDSN code XE, 2018–2019) on ∼40 Ma Pacific seafloor formed at the fast-spreading Pacific–Farallon ridge, at water depths of 4386–4824 m. The stations carry Nanometrics Trillium 240 seismometers and differential pressure gauges, as do the RHUM-RUM stations of the earlier study. After removing clipped and dead data, earthquakes, tilt and horizontal-coherent noise, and calibrating each pressure gauge on Rayleigh waves of $M \geq 7$ earthquakes (gain factors 0.70–1.24, ±1.4 per cent between events), we measure the normalized compliance between 4.9 and 14.0 mHz at 20 stations from 5038 35-min windows per station (median), with coherence above 0.9 and month-to-month variations of a few per cent. A Bayesian inversion (delayed-rejection adaptive Metropolis with parallel tempering, free crustal Vp/Vs, pressure gain sampled within ±5 per cent) fits every station (median $\chi^2/N = 0.36$). Below 79–210 m of sediment, consistent with CRUST1.0, the network-pooled shear velocity rises from 2.37 km s^{-1} at 0.3 km to 2.94 km s^{-1} at 1 km and 3.23 km s^{-1} at 2 km below the seafloor, slower than CRUST1.0 in the upper crust. Lower-crustal Vs depends on the Vp/Vs assumption (3.54 with Vp/Vs free, 4.05 km s^{-1} with Vp/Vs = 1.85). Converted with the relation used for the Indian Ocean ridges, the upper-crustal velocities would imply 63 (27–91) per cent serpentinization at 1 km; on magmatic crust formed at a fast-spreading ridge they instead record the porosity of the extrusive section. This normal-crust reference shows that compliance-derived Vs alone cannot identify serpentinite, and it calibrates the interpretation of the low velocities found at the South-West Indian Ridge.

Key words: Oceanic crust; Seafloor compliance; Inversion; Pacific Ocean; Monte Carlo methods.

1 INTRODUCTION

Seafloor compliance, the transfer function between vertical seafloor displacement and the pressure of infragravity waves, constrains the shear velocity (Vs) of the upper few kilometres beneath a single broad-band ocean-bottom seismometer (OBS) equipped with a differential pressure gauge (DPG) (Crawford et al. 1991, 1998). Aminian et al. (2025) measured it at eight RHUM-RUM stations in the Indian Ocean and interpreted very low upper-crustal Vs beneath the ultra-slow South-West Indian Ridge (SWIR) as serpentinized mantle-derived peridotite.

Two questions follow. Is the processing reproducible on another network without manual tuning? And what does the same analysis give on crust that is *not* expected to contain serpentinite? The Young ORCA experiment (Eilon et al. 2022) answers both. Its 30 broad-band OBS were deployed for about 13 months in 2018–2019 on ∼40 Ma Pacific lithosphere, formed at the fast-spreading Pacific–Farallon ridge, a setting where the crust is fully magmatic and layered in the classical way (e.g. Christeson et al. 2019).

This report applies ComPy 2.4.0, which now runs from the raw archive to compliance, Vs and serpentinization unattended, to the whole network, using the method of Aminian et al. (2025) with the corrections of ComPy 2.2–2.4 (Section 4). We present the compliance functions, the Vs models and their resolution, and discuss what they imply for the serpentinization interpretation.

2 THEORY

The vertical compliance $\zeta(\omega)$ is the ratio of the spectra of vertical displacement and vertical stress at the seafloor (Sorrells & Goforth 1973). We use the normalized compliance $\eta(\omega) = k(\omega)\zeta(\omega)$, with k the infragravity wavenumber from $\omega^2 = gk\tanh(kH)$ for water depth H. The measured compliance is

$$\hat{\eta}(\omega) = k\,\frac{\omega^2\,|\hat{S}_{pz}|/\hat{S}_{pp} + a_w/p}{\omega^2 + 2g/R}, \qquad (1)$$

where $\hat{S}_{pz}$ and $\hat{S}_{pp}$ are the cross- and pressure spectra stacked over the selected windows, $a_w/p = 2\pi(G/g)\,e^{-kH}\cosh kH$ is the gravitational attraction of the passing water mass per unit pressure, and $2g/R$ the free-air gradient (Crawford et al. 1998). The ratio $|\hat{S}_{pz}|/\hat{S}_{pp}$ equals the coherence times $\sqrt{\hat{S}_{zz}/\hat{S}_{pp}}$ used in the paper's eq. (6). The wave-attraction term is deterministic and is not multiplied by the coherence: earlier ComPy versions did so (2.0–2.1) or dropped it (before 2.0), which biased η low at the lowest frequencies. The

random error follows Bendat & Piersol (1993),

$$\epsilon[\hat{\eta}] = \frac{\sqrt{1-\hat{\gamma}^2}}{|\hat{\gamma}|\sqrt{2n_d}}\,|\hat{\eta}|, \tag{2}$$

with n_d the number of independent spectral averages over all accepted windows.

The forward model is the minor-vector propagator of Crawford et al. (1991) for a layered elastic half-space. The likelihood is Gaussian with a full data covariance $\mathbf{C}$,

$$L(\mathbf{m}) \propto \exp\!\left[-\tfrac{1}{2}\,(\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}(\mathbf{m}))^{\mathsf{T}}\mathbf{C}^{-1}(\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}(\mathbf{m}))\right], \tag{3}$$

where $\mathbf{C}$ sums the random error, the covariance of monthly stacks (which measures non-stationarity and correlates neighbouring frequencies), and a 1 per cent floor for 2-D structure. The pressure gain is a parameter of the model, so the 5 per cent calibration uncertainty is shared by all frequencies and is not counted once per frequency.

Serpentinization β is converted from Vs with the linear relation of Miller & Christensen (1997) used in Aminian et al. (2025, Fig. 10):

$$\beta = 100\,\frac{4.2 - V_s}{4.2 - 2.2}\ \text{per cent} \quad (V_s \text{ in km s}^{-1}), \tag{4}$$

clipped to 0–100 and evaluated on every posterior draw.

3 DATA SET

Young ORCA (Gaherty et al. 2019; Eilon et al. 2022) deployed 30 Scripps broad-band OBS between 4°S and 8°S, 130°–136°W (Fig. 1), in water 4386–4824 m deep. Each carries a Trillium 240 seismometer and a DPG, recorded at 50 Hz. The data come from the BRUIT-FM archive (SDS, 587 GB for XE), with responses from the EarthScope/IRIS DMC. Because reading the archive disk in parallel was slow, the raw files of most stations were first copied to local disk station by station; the processing is identical.

Three stations were not processed: CC02, whose pressure channel does not work, and CC10 and CC12, which recorded for 50 and 17 days. Of the 27 processed stations, 3 have no 13-h span in which all four channels are usable (EC02, EE02, EE03; the seismometer is clipped or dead most of the time, Section 4.1), at WW02 no window passes the quality gates, and 3 do not reach the 500 windows required by Aminian et al. (2025): EC03 (274), EC04 (5), WC01 (3). Table 1 lists the 20 stations with a compliance measurement; 20 of them are inverted.

4 DATA PROCESSING, COMPLIANCE MEASUREMENT AND INVERSION

The processing follows the flowchart of Aminian et al. (2025, Fig. 2), run unattended by `ComPy/pipeline`.

4.1 Pre-processing

Decimation, clipping and responses. The raw data are decimated to 1.25 Hz with FIR filters (factors 5, 4, 2).

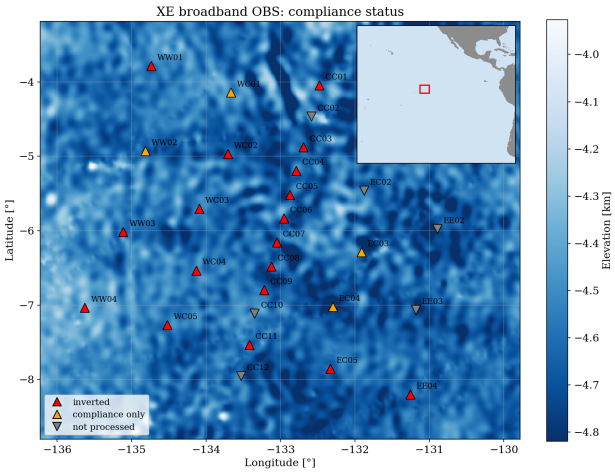


Figure 1. Young ORCA (XE) stations on bathymetry (ETOPO1). Red: compliance measured and inverted; orange: compliance measured but rejected or not inverted; grey: not processed or no usable data. Inset: location in the Pacific.

Many Young ORCA seismometers sat at full scale (2^{23} counts) for weeks, presumably while the sensor was off level, and some channels went flat. Following the paper's rule that windows with a saturated component are discarded, every 10-min block in which any channel has a sample within 1 per cent of full scale, or a standard deviation below 5 counts, is cut from all channels, with 5 minutes on each side, before decimation. This removed 28 per cent of the record at the processed stations. Responses are removed after decimation without a water level and with an explicit pre-filter (0.8–600 mHz corners), giving displacement and pressure.

Earthquakes and local events. Spans after $M \geq 5.85$ earthquakes in the USGS catalogue (1.5 d per magnitude unit; Deen et al. 2017) and local events detected with a recursive STA/LTA trigger (0.1–0.5 Hz) are excluded from the spectral estimates.

Glitches. The hourly mass-position glitch of the RHUM-RUM instruments is absent here: folding 15-day blocks at trial periods around 1 h gave fold strengths of 1.15 (median) and 1.76 (maximum) at six stations, below the 2.0 needed to subtract a template, so the step was switched off for the rest of the network.

Tilt and coherent noise. Hourly tilt angles that minimize the variance of the vertical (Harmon et al. 2022) are median-smoothed over 24 h and applied per sample, so the rotation adds no steps at hour boundaries (Fig. 3). The remaining horizontal-coherent noise is then removed from the vertical with transfer functions estimated over ≤24 h windows (Crawford & Webb 2000). Fig. 2 shows the vertical acceleration PSD at CC11 before and after cleaning; the compliance bump near 0.01 Hz stands at -158 dB.

4.2 Pressure gauge calibration

Each DPG is calibrated in situ on the Rayleigh waves of $M \geq 7$ earthquakes (Zha & Webb 2016; Aminian et

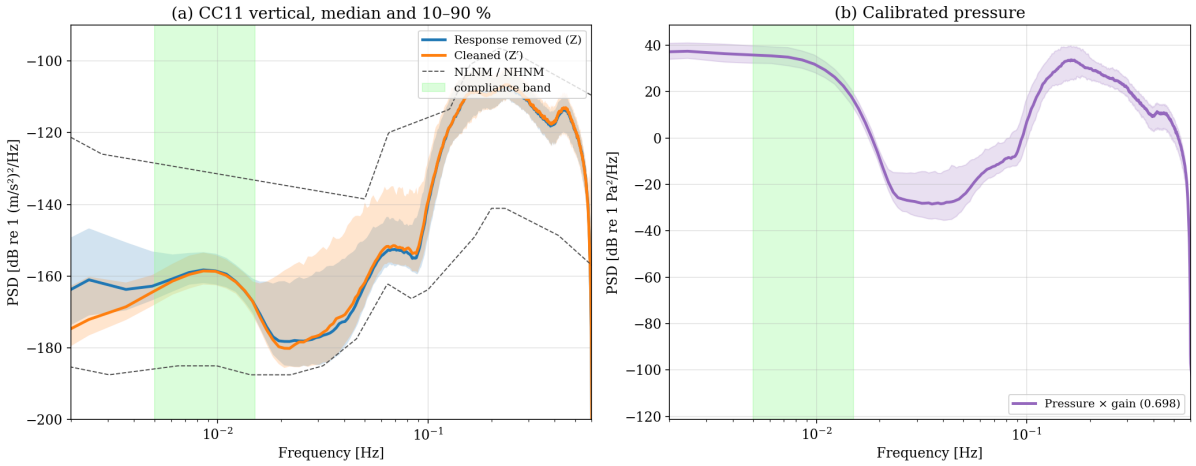


Figure 2. (a) Vertical acceleration PSD at CC11 after response removal (blue) and after tilt and coherent-noise removal (orange); median and 10–90 per cent range of 1-h windows; green: compliance band. (b) Calibrated pressure PSD.

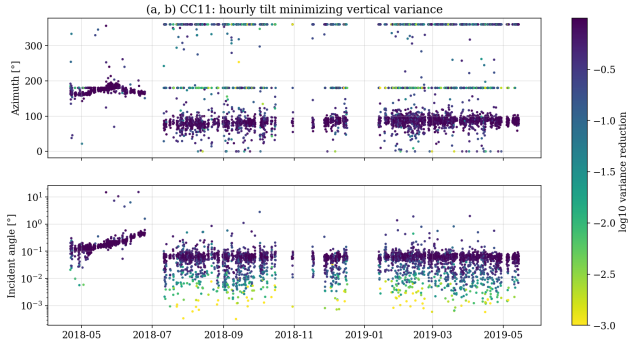


Figure 3. Hourly tilt azimuth (a) and incidence angle (b) at CC11, coloured by the logarithm of the variance reduction.

al. 2025): the measured pressure-to-acceleration ratio, normalized by $\rho_w H$, is fitted to its theoretical value between 0.02 and 0.06 Hz for events whose magnitude-squared coherence exceeds 0.97 (Fig. 4). Gains were obtained at 18 of the 20 stations with a compliance measurement, from 1–12 events each, and range from 0.70–1.24 with an event-to-event spread of 0.1–3.2 per cent. Where the BRUIT-FM microseism calibration exists, the ratio of the two gains is 0.986 (0.835–1.029) at 13 of 14 stations; they disagree at EC01 (ComPy 0.93, BRUIT-FM 3.51), where one of the two methods failed. Gains from a single event (Table 1) are less certain than the ±5 per cent prior assumes. No event passed the coherence test at CC01 and CC09; these stations were kept with the nominal response and a gain sampled over 0.6–1.4, and are excluded from the pooled results.

4.3 Compliance measurement

Compliance is computed on non-overlapping 35-min windows. A window is kept if, in the compliance band, its median coherence is at least 0.9 (0.8 if fewer than 500 windows pass), the mean calibrated pressure PSD is 10–50 dB re 1 $\mathrm{Pa}^2\,\mathrm{Hz}^{-1}$, and the mean vertical acceleration PSD is between −170 and −150 dB re 1 $(\mathrm{m\,s}^{-2})^2\,\mathrm{Hz}^{-1}$ (Aminian et al. 2025). The band is the widest run of frequencies where the mean coherence of the kept windows reaches the threshold, below the infragravity cut-off $\sqrt{g/2\pi H}$. Spectra of the kept windows are stacked before the ratio is formed (eq. 1). Fig. 5 shows one month at CC11; the coherence is high whenever the vertical noise is low. Fig. 6 shows the selected windows and the stacked compliance. Monthly stacks with the same band test the temporal stability.

4.4 Shear-velocity inversion

The inversion follows the parameterization of Zha & Webb (2016) as implemented in ComPy 2.3. A sediment layer of free thickness (0–1000 m) has Hamilton (1976, 1979) Vs and Vp profiles. Below it, crustal layers 100, 200, 400, 800 and 1600 m thick reach the CRUST1.0 Moho at the station, above a mantle half-space. Crustal Vs is uniform on 0.5–4.7 $\mathrm{km\,s}^{-1}$ with a hierarchical second-difference smoothing prior. Crustal Vp/Vs is free on 1.7–2.4 with density from Christensen & Shaw (1970), and the pressure gain is sampled within ±5 per cent of the Rayleigh value. Eight chains of delayed-rejection adaptive Metropolis (Haario et al. 2006), each a parallel-tempering ladder of eight temperatures, run 80 000 iterations (30 000 burn-in) from independent prior draws; a prior-only run shows what the data add. Convergence is assessed with the rank-normalized $\hat{R}$ and effective sample size (Vehtari et al. 2021). The variant with Vp/Vs fixed at 1.85 brackets the effect of that assumption. This differs from the published workflow, whose code is audited in `ComPy/docs/INVERSION.md`.

5 RESULTS

5.1 Compliance functions

Table 1 and Fig. 7 summarize the measurements. At the 18 calibrated stations the band is 4.9–14.0 mHz, the median coherence 0.902–0.970, and the random error about 0.1 per cent. The functions are remarkably similar across the array: η at the lowest frequency is 0.016–0.028 GPa^{-1} and rises monotonically

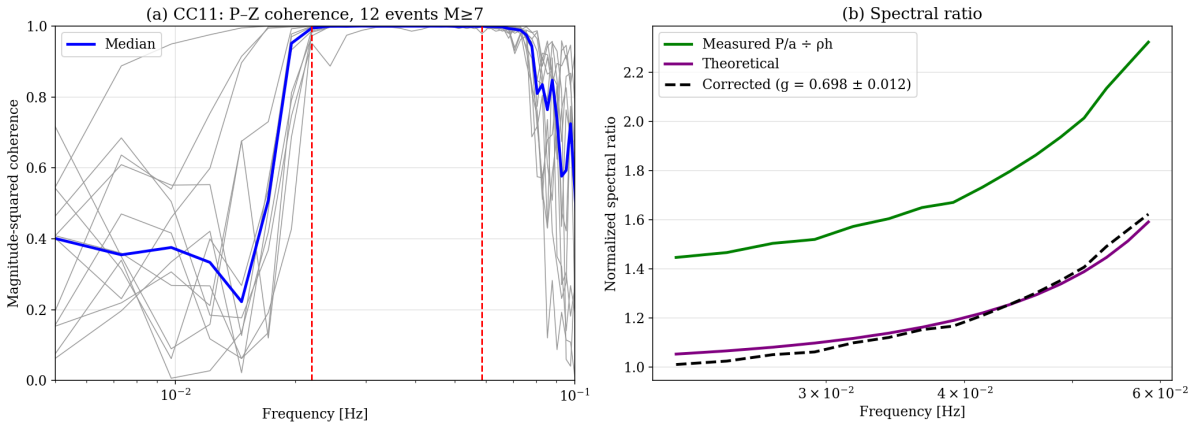


Figure 4. DPG calibration at CC11. (a) Pressure–vertical coherence of the accepted events and its median; dashed: fitted band. (b) Measured, theoretical and corrected normalized pressure/acceleration ratio.

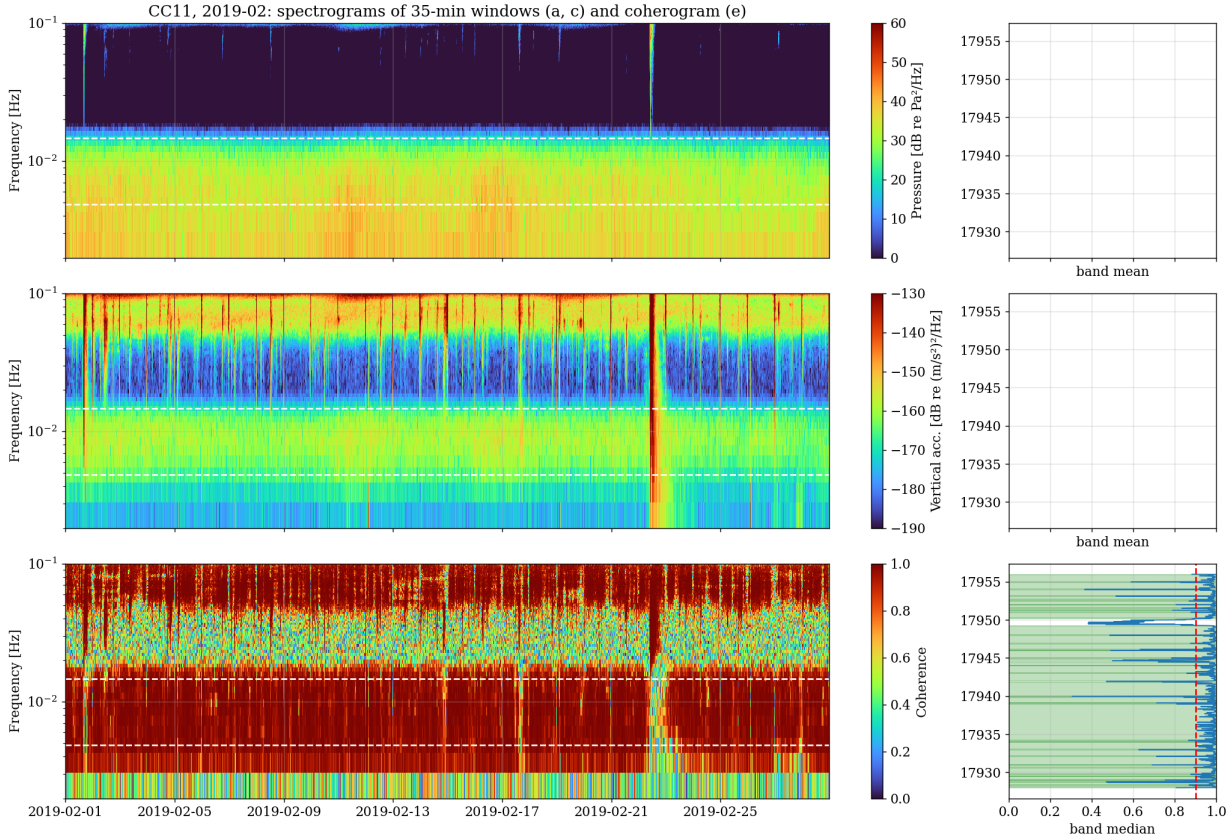


Figure 5. One month at CC11: spectrograms of calibrated pressure (a) and corrected vertical acceleration (b) for 35-min windows, coherogram (c), and their band averages (right). Dashed lines: compliance band; shading: windows above the coherence threshold.

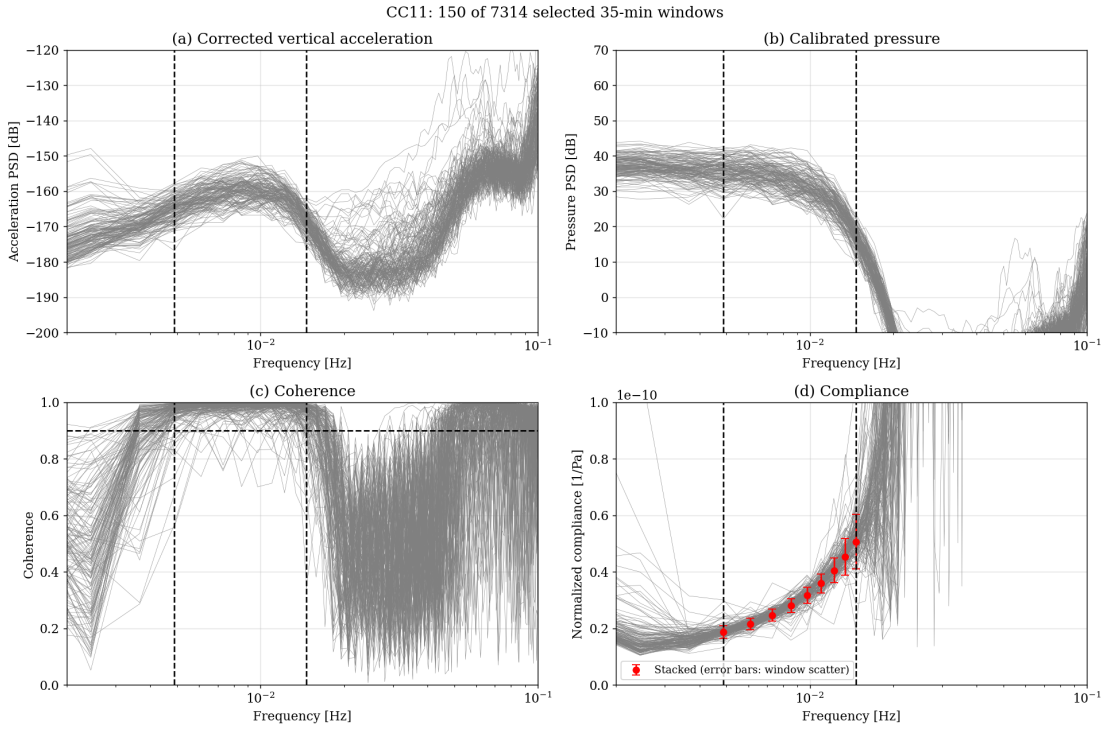


Figure 6. Selected windows at CC11 (grey, a random subset): (a) vertical acceleration, (b) pressure, (c) coherence, (d) compliance, with the stacked estimate (red; error bars: window scatter). Dashed: band and coherence threshold.

with frequency, the signature of Vs increasing with depth. Monthly stacks agree with the full stack to within 1–3 per cent at most stations; CC07 is the exception (-25 to +9 per cent), with a drift that the Rayleigh-wave gain does not show and which therefore enters its covariance as non-stationarity.

5.2 Shear-velocity profiles

All inversions converged ($\hat{R} \leq 1.015$) and fit to χ^2 below the number of data. Fig. 8 shows the posteriors and Table 2 the values at fixed depths. The sediment is 79–210 m thick, in line with CRUST1.0 (70–190 m). Below it, Vs rises steeply to 2.35–2.86 km s^{-1} at 0.5 km and 2.67–3.20 km s^{-1} at 1 km, then more slowly to 3.01–3.58 km s^{-1} at 2–3 km. The posterior is 0.36 times as wide as the prior at 1 km and 0.21 at 2 km, so these depths are resolved. Half of the sensitivity of the lowest frequency lies above 3.5 km and 90 per cent above 12.3 km. Below about 3 km the posterior width reflects the layer geometry as much as the data.

5.3 Vp/Vs and the lower crust

The crustal Vp/Vs posteriors lean towards high values (medians 2.07–2.26 on a 1.7–2.4 prior) but remain wide. Compliance depends on both Lamé parameters, so a higher Vp/Vs is traded for a lower Vs. With Vp/Vs fixed at 1.85 the upper 2 km barely change, while Vs at 4 km rises from 3.54 to 4.05 km s^{-1} (network median), close to CRUST1.0 (4.05 km s^{-1}). Both models fit. Lower-crustal Vs is therefore set by the Vp/Vs assumption, whereas upper-crustal Vs is a robust result.

5.4 Serpentinization

Applied to the pooled posterior, eq. (4) gives 78 (20–100) at 0.5 km, 63 (27–91) at 1 km and 49 (25–67) per cent at 2 km below the seafloor (median and 5–95 per cent range; Fig. 9c).

6 DISCUSSION

A consistent normal-crust signal. The Young ORCA compliance is uniform across a 600 km by 500 km array, and the Vs profiles agree between stations to within their uncertainties. This is expected from crust formed at a fast-spreading ridge, where crustal thickness and layering vary little (e.g. Christeson et al. 2019). The velocity structure is the classical one: a thin pelagic sediment cover; a steep gradient through the extrusive section (layer 2A–2B), where Vs climbs from ∼2.3 to ∼3 km s^{-1} in the first kilometre as porosity and crack density decrease (Carlson 2014); and a gentler gradient in the dikes and gabbros. Upper-crustal Vs is lower than in CRUST1.0, which averages refraction models and smooths the steep gradient at the top of the crust.

What this means for serpentinization. Fed through the Miller & Christensen (1997) relation, the upper-crustal velocities at Young ORCA would read as 63 per cent serpentinization at 1 km, comparable with the values now obtained beneath the SWIR stations of RHUM-RUM (30–55 per cent at 2 km with the corrected relation and models; ComPy 2.4 report). Serpentinite is not expected in this magmatic crust; the low Vs is the porosity of the extrusives. Vs from compliance alone therefore cannot distinguish serpentinized peridotite from porous basalt. The SWIR interpretation

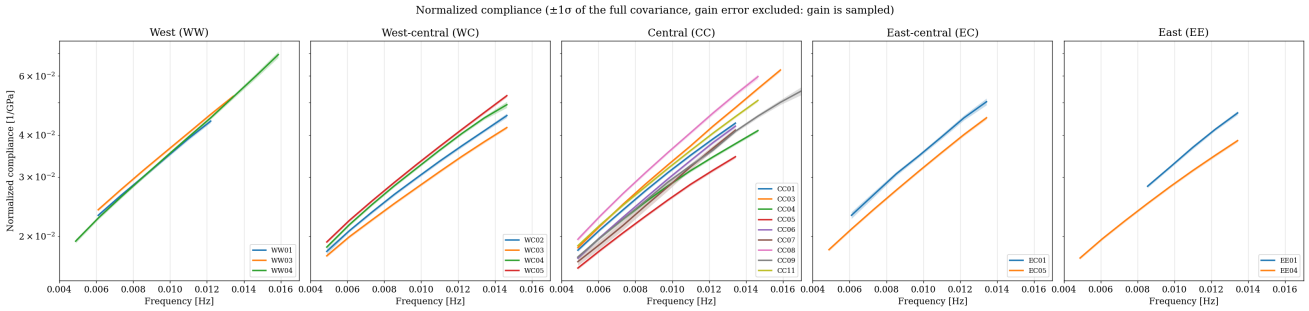


Figure 7. Normalized compliance by sub-array, west to east; shading: $\pm 1\sigma$ from the full covariance.

Table 1. Stations with a compliance measurement. Gain: ComPy Rayleigh-wave calibration (event standard deviation, number of events); BRUIT-FM: independent microseism calibration where available. Frequency: compliance band. Threshold: coherence gate. Coherence: band median of the stack.

Station	Lat. [°]	Lon. [°]	Depth [m]	Gain (ComPy)	Gain (BRUIT-FM)	Band [mHz]	Thr.	Windows	Coh.
CC01	-4.0458	-132.4745	4768	–	–	4.9–13.4	0.9	2781	0.968
CC03	-4.8816	-132.6890	4739	0.763 (1)	0.913	4.9–15.9	0.9	1149	0.965
CC04	-5.1937	-132.7900	4615	0.804 ± 0.010 (7)	0.803	4.9–14.6	0.9	7024	0.967
CC05	-5.5174	-132.8730	4798	1.060 ± 0.016 (10)	1.146	4.9–13.4	0.9	7602	0.960
CC06	-5.8385	-132.9518	4824	0.939 ± 0.019 (6)	0.965	4.9–13.4	0.9	5002	0.964
CC07	-6.1588	-133.0464	4628	0.912 ± 0.013 (9)	–	4.9–13.4	0.9	5773	0.942
CC08	-6.4816	-133.1233	4745	0.839 ± 0.001 (5)	–	4.9–14.6	0.9	3936	0.964
CC09	-6.7995	-133.2164	4615	–	–	4.9–17.1	0.8	501	0.957
CC11	-7.5368	-133.4166	4636	0.698 ± 0.012 (12)	0.702	4.9–14.6	0.9	7314	0.970
EC01	-4.7261	-131.4852	4576	0.927 (1)	3.509	6.1–13.4	0.8	545	0.902
EC05	-7.8562	-132.3298	4587	0.840 ± 0.012 (5)	0.824	4.9–13.4	0.9	4093	0.961
EE01	-4.9470	-130.3814	4645	0.872 ± 0.014 (10)	0.893	8.5–13.4	0.9	3207	0.926
EE04	-8.2043	-131.2551	4683	1.146 ± 0.015 (12)	1.159	4.9–13.4	0.9	7316	0.965
WC02	-4.9726	-133.7054	4619	1.242 ± 0.009 (6)	1.259	4.9–14.6	0.9	5074	0.967
WC03	-5.7077	-134.0913	4483	0.779 ± 0.008 (9)	0.778	4.9–14.6	0.9	8572	0.968
WC04	-6.5352	-134.1305	4534	0.865 ± 0.023 (6)	0.840	4.9–14.6	0.9	7145	0.965
WC05	-7.2694	-134.5177	4478	0.898 ± 0.015 (9)	0.992	4.9–14.6	0.9	7222	0.961
WW01	-3.7852	-134.7370	4534	0.821 ± 0.026 (4)	0.925	6.1–12.2	0.9	1796	0.945
WW03	-6.0155	-135.1120	4386	0.873 ± 0.011 (12)	–	6.1–13.4	0.9	5958	0.959
WW04	-7.0385	-135.6272	4491	0.911 ± 0.003 (3)	–	4.9–15.9	0.9	2736	0.964

Table 2. Inversion results: fit (χ^2/N), largest $\hat{R}$, sediment thickness and crustal Vp/Vs (posterior medians), Vs [km s^{-1}] as median (5–95 per cent) at depths below the seafloor, and Vs at 4 km with Vp/Vs = 1.85. *Gain not calibrated (sampled over 0.6–1.4).

Station	χ^2/N	$\hat{R}$	Sed. [m]	Vp/Vs	Vs 0.5 km	Vs 1 km	Vs 2 km	Vs 4 km	Vs 4 km (1.85)
CC01*	2.9/8	1.015	147	2.16	2.49 (1.79–3.70)	2.72 (2.14–3.48)	2.93 (2.51–3.42)	3.21 (2.74–3.80)	3.98
CC03	3.6/10	1.011	166	2.24	2.64 (2.10–3.38)	2.94 (2.61–3.30)	3.25 (3.04–3.49)	3.55 (3.34–3.92)	4.22
CC04	3.1/9	1.006	81	2.23	2.64 (2.11–3.48)	2.85 (2.46–3.33)	3.09 (2.86–3.33)	3.35 (3.19–3.63)	3.89
CC05	3.0/8	1.006	79	2.21	2.80 (2.18–3.85)	3.02 (2.52–3.71)	3.28 (2.93–3.62)	3.56 (3.37–3.89)	4.09
CC06	3.0/8	1.007	118	2.26	2.58 (1.98–3.71)	2.88 (2.38–3.59)	3.22 (2.84–3.56)	3.58 (3.39–3.95)	4.31
CC07	2.8/8	1.009	137	2.17	2.83 (2.06–4.09)	3.20 (2.48–4.02)	3.58 (3.07–4.10)	3.99 (3.60–4.55)	4.44
CC08	3.2/9	1.015	153	2.23	2.35 (1.79–3.40)	2.70 (2.23–3.23)	3.08 (2.78–3.39)	3.48 (3.24–3.91)	4.16
CC09*	5.4/11	1.010	85	2.12	2.27 (1.81–3.01)	2.64 (2.20–3.12)	3.08 (2.66–3.49)	3.52 (3.02–4.07)	4.17
CC11	3.2/9	1.009	121	2.17	2.53 (1.94–3.43)	2.85 (2.39–3.37)	3.17 (2.89–3.48)	3.50 (3.26–3.98)	3.98
EC01	3.0/7	1.006	165	2.12	2.63 (1.90–3.99)	2.93 (2.25–3.72)	3.22 (2.70–3.68)	3.52 (3.16–4.19)	3.89
EC05	2.9/8	1.010	133	2.20	2.61 (1.94–3.74)	2.90 (2.35–3.58)	3.24 (2.86–3.58)	3.58 (3.34–4.01)	4.18
EE01	3.0/5	1.010	138	2.07	2.63 (1.92–3.94)	3.00 (2.29–3.83)	3.36 (2.78–3.87)	3.78 (3.39–4.41)	4.11
EE04	2.8/8	1.006	108	2.18	2.86 (2.12–3.90)	3.06 (2.49–3.76)	3.28 (2.93–3.64)	3.48 (3.27–3.86)	3.96
WC02	3.0/9	1.006	112	2.16	2.72 (2.09–3.65)	2.96 (2.48–3.53)	3.20 (2.92–3.48)	3.43 (3.21–3.83)	3.88
WC03	3.4/9	1.005	95	2.12	2.77 (2.12–3.72)	3.04 (2.55–3.59)	3.31 (3.01–3.62)	3.58 (3.33–4.02)	3.96
WC04	3.7/9	1.009	100	2.24	2.39 (1.90–3.39)	2.67 (2.24–3.23)	3.01 (2.73–3.29)	3.39 (3.19–3.74)	4.06
WC05	3.1/9	1.006	139	2.16	2.61 (2.00–3.51)	2.90 (2.46–3.40)	3.18 (2.91–3.47)	3.46 (3.22–3.92)	3.94
WW01	2.5/6	1.009	179	2.09	2.76 (1.96–4.13)	3.08 (2.37–3.89)	3.34 (2.81–3.88)	3.60 (3.24–4.23)	3.94
WW03	2.7/7	1.009	170	2.11	2.58 (1.89–3.89)	2.91 (2.32–3.65)	3.23 (2.82–3.67)	3.53 (3.20–4.17)	3.92
WW04	3.7/10	1.006	210	2.23	2.76 (2.15–3.84)	3.13 (2.69–3.63)	3.35 (3.10–3.66)	3.57 (3.33–4.02)	4.19

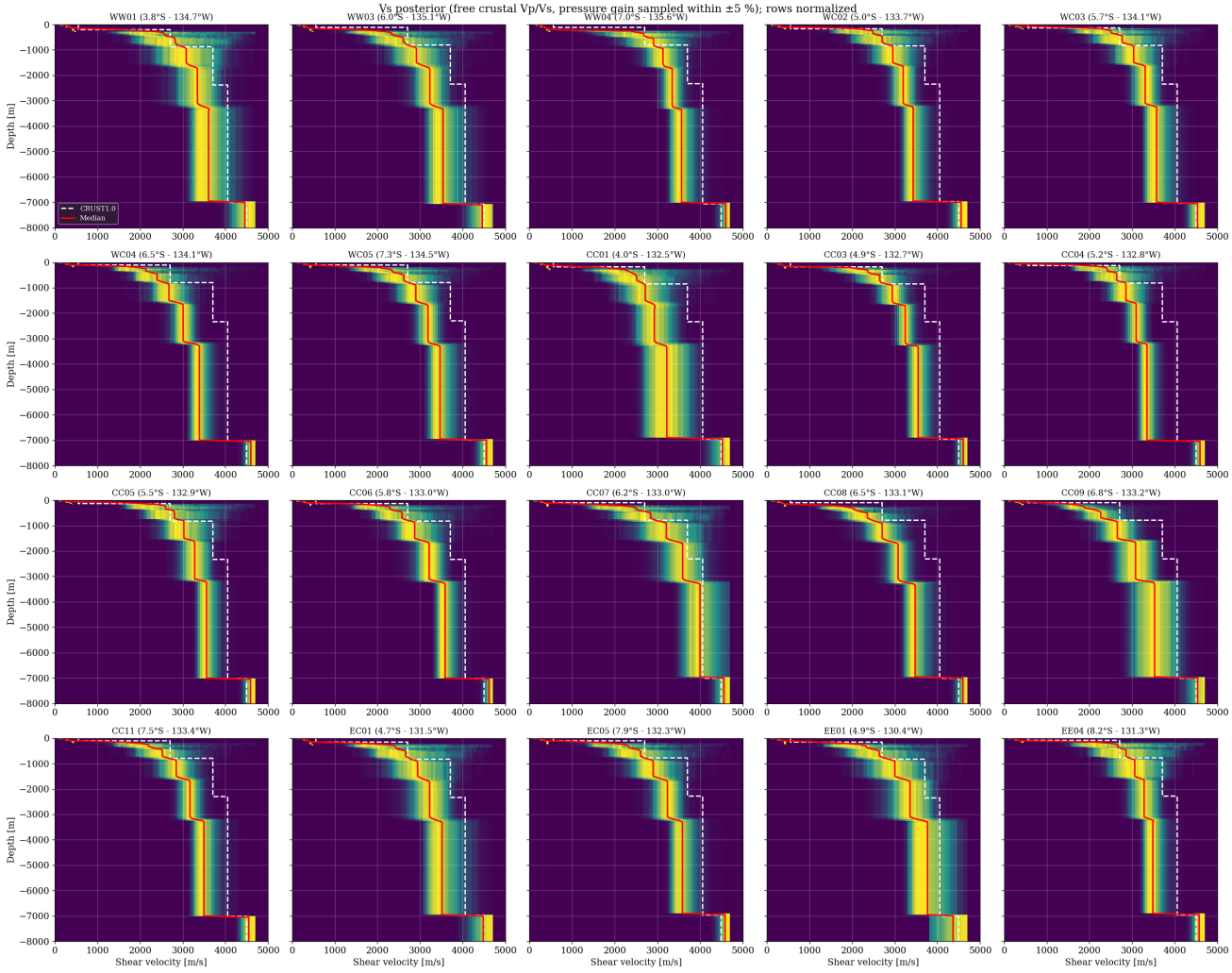


Figure 8. Vs posterior heat maps (rows normalized). Red: median; white dashed: CRUST1.0 at the station. Depths are true depths below the seafloor.

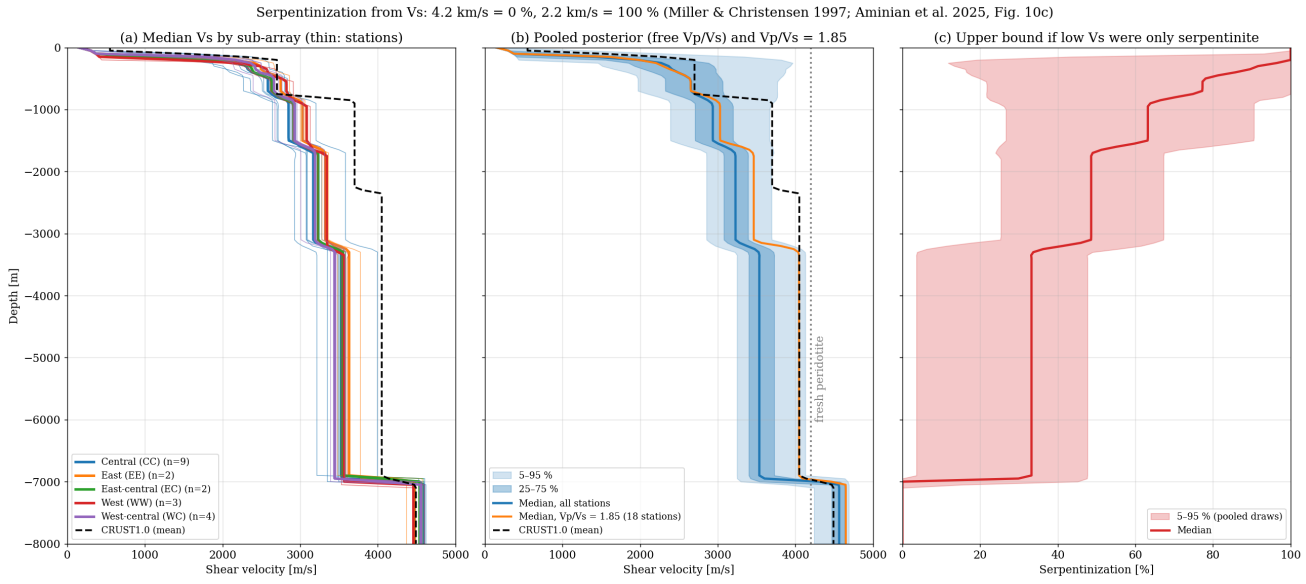


Figure 9. (a) Median Vs by sub-array (thin lines: stations) and the mean CRUST1.0 profile. (b) Pooled posterior of the calibrated stations with Vp/Vs free (blue) and the median with Vp/Vs = 1.85 (orange). (c) Serpentinization from eq. (4) applied to the pooled draws: the value that *would* follow if the crust were peridotite.

has to rest on the geological setting (exhumed mantle in oblique, magma-poor sections; Sauter & Cannat 2010) and, where possible, on independent constraints: Vp/Vs, which is high in serpentinite ($\geq$2.0; Christensen 2004) but is poorly resolved by compliance (Section 5.3), density, or active-source data. A useful diagnostic is the *difference* between ridge and normal crust at equal depth. At 2 km below the seafloor, Young ORCA gives 3.23 km s^{-1}; the corrected models of RR36 and RR40 give 3.09 and 3.56 km s^{-1} and RR52 (CIR) 3.59 km s^{-1}. None of them is slower than the Young ORCA range (3.01–3.58 km s^{-1} at 2 km), so at that depth the ridge stations are not distinguishable from normal crust by Vs alone.

Limitations. (i) The upper bound of the Vp/Vs prior is reached by several posteriors, so a wider prior could lower lower-crustal Vs further. (ii) The compliance of CC07 drifts slowly; the drift widens its covariance and posterior. CC01 and CC09 have no Rayleigh-wave calibration (gain sampled over 0.6–1.4); their profiles are shown but not pooled. EC01 and CC03 rest on a single calibration event. (iii) A 1-D model ignores the abyssal-hill topography, partly covered by the 1 per cent covariance floor (Hulme et al. 2005). (iv) Stations with long clipped spans have short records.

7 CONCLUSIONS

ComPy 2.4.0 processed a second broad-band OBS network from the raw archive to compliance, shear velocity and serpentinization without manual intervention, after adding the removal of clipped data that this network required. At 20 Young ORCA stations it measures compliance with coherence above 0.9 over 4.9–14.0 mHz, and its Rayleigh-wave DPG calibration agrees with an independent microseism calibration. The inverted crust of $\sim$40 Ma Pacific lithosphere has 79–210 m of sediment and upper-crustal Vs of 2.35–2.86 km s^{-1} at 0.5 km. Lower-crustal Vs depends on Vp/Vs. These normal-crust velocities would translate into substantial "serpentinization" if read like the SWIR results, so compliance-derived Vs must be combined with geological and independent geophysical constraints before it is interpreted as alteration.

DATA AND SOFTWARE

Young ORCA data (FDSN network XE, 2018–2019) are distributed by the EarthScope Consortium (IRIS DMC). ComPy 2.4.0, including the pipeline and the scripts of this run, is at https://github.com/MohammadAmin-Aminian/ComPy. Run folder: `BRUIT-FM/compliance_xe_2026-10-29`.

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